CHAPTER 7

Primitives and differential equations

S52, S53, S55, S58

Before you start · Derivation and models

Without a calculator, in six minutes. Respond briefly. Mobilize derivative, exponential and units. Then choose the targeted pick-up indicated under the questions.

1. Derivating $F(x) = x^2$.

2. Find an affine function whose derivative is 3.

3. Derive 3 e^{2x}.

4. In $f(t) = 80e^{-0,1t}$, calculate $f(0)$.

5. Convert 90 min to hours.

6. What is worth e^0 ?

Responses

$F' = 2x$; $3x + C$; $6e^{2x}$; 80 ; 1,5 h ; 1.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Retake the derivative x squared. **2.** Write mx+b and drift to determine m. **3.** Identify the factor 2 from the composition; R36. **4.** Substitute t=0 before calculating the exponential. **5.** Divide minutes by 60; R67. **6.** Retrieve the value of the original exp. Questions R serve as support for review with the teacher, even if their collective ritual is planned later.

Objectives

Calculate antiderivatives; solve a differential equation with initial condition; interpret an approximation of Euler.

Course notes

Primitives over an interval

It is admitted at this stage that any function continues over an interval admits antiderivatives; the connection with Chapter 8 will be prepared for this purpose. f over an interval I is a detritable function F such that $F' = f$ on I. If F is a antiderivative, all antiderivatives on I are $F + C$, $C \in \mathbb{R}$.

Demonstration. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval hypothesis is essential: on a disjunct domain, the constants may differ according to the intervals.

Table of antiderivatives · inverse reading of derivatives

$f(x)$	$F(x)$
x^n $(n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ over an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\sqrt{x}$ $(x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two lines will be reinvested After the study of trigonometry.

Recognition of membership

For a function u Drifting, antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln \lvert u \rvert$, on intervals where expressions have a meaning. Linearity allows to process sums.

Example. A antiderivative of 6 $x(x^2+1)^2$ is $(x^2+1)^3$; a antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation connects an unknown function to its derivatives. over a specified interval.

Common solutions

$y' = ay$ a for solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ a for solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a particular known solution of $y' = ay + f(x)$, solutions are $y = g + Ce^{ax}$.

Demonstration for $y' = ay$**.** If y is a solution, $z(x) = e^{-ax}y(x)$ checks $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtract the constant solution $-\frac{b}{a}$ the same equation. For a second variable member, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$then $C = -2$. Solution 3 $-2e^{-2x}$ growth to balance 3 when $x \to +\infty$.

Method of production

For $y' = ay + b$, with one step $h > 0$, we're approaching $y(t_k)$ by Y_k by $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. It is an approximation using tangent, not equality on the exact solution. A smaller step can improve approximation; numerically control error on a solution known.

Dive backwards and model an evolution

7.1 Recognize and control a antiderivative

First use: S52.

Definition · Primitive over an interval

A F function is a antiderivative f on I if F is driftable on I and $F' = f$ I. Interval is part of the data. Finding a antiderivative is an inverse calculation; the derivation allows to Check it out.

Example Solved · Correct a Factor

For $f(x) = 3e^{2x}$, $F(x) = \frac{3}{2}e^{2x}$ is appropriate because $F'(x) = \frac{3}{2} \times 2e^{2x} = 3e^{2x}$. For $g(x) = 2x/(1+x^2)$ on $\mathbb{R}$, the antiderivative $G(x) = \ln(1+x^2)$ uses a positive argument. All antiderivatives obtain by adding a constant.

A1 · Immediate application Determine a antiderivative of 4 $x^3 - 2x + 1$, of e^{-3x} and 1 $/(2x+1)$ on a suitable interval. Check by bypass.

7.2. A constant, a vertical translation

First use: S52.

Demonstration. If $F' = G' = f$ over interval I, then $(F-G)' = 0$ on I. A derivative function zero over an interval is constant; therefore $F - G = C$. On a domain consisting of several intervals Disjointed, the constant can be different over each interval. □

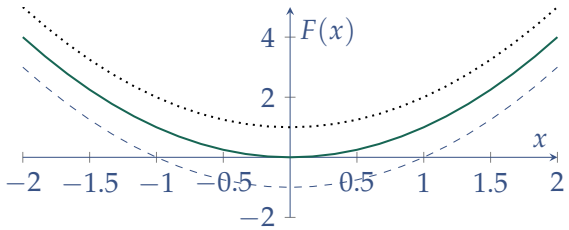


Figure 7.1. Three antiderivatives of $2x$: On the same apses, the tangents have the same slopes.

Solved Example · A value sets the translation

The antiderivatives of $2x$ are $x^2 + C$. Imposing F(0)=1 selects $F(x) = x^2 - 1$. The graph is Translating a unit downwards without changing the slopes.

A2 · Partial guidance Determine the antiderivative of $2x$ which is worth 3 in 1. Then explain why all the antiderivatives of $1/x$ on $\mathbb{R} \setminus \{0\}$ are not described by a single constant common to both Half straight.

7.3. Unknown is a function

First use: S55.

Ownership · Equation $y' = ay$

For $a \in \mathbb{R}$, solutions on $\mathbb{R}$ are $y(t) = Ce^{at}$, $C \in \mathbb{R}$. An initial value $y(0) = y_0$ imposes $C = y_0$.

Demonstration. A function of this form is well solution. Conversely, if it is solution, $z(t) = y(t)e^{-at}$ a for derivative $(y' - ay)e^{-at} = 0$, so is constant over the interval. $y(t) = Ce^{at}$. The evidence checks the solutions and shows that none is missing. □

Example resolved · A condition in another instant

For $y' = 3y$ and y(1)=2, $y(t) = Ce^{3t}$then $Ce^3 = 2$. Therefore $C = 2e^{-3}$ and $y(t) = 2e^{3(t-1)}$. The constant is equal to the value imposed only if it is given in 0.

A3 · Immediate application Solve $y' = -2y$ with $y(0) = 5$. Is the number 0 a solution possible differential equation alone ? Of this problem with initial condition ?

7.4 Subtract a known solution

First use: S55 and S58.

Example resolved · Second variable member

For $y' = -y + 2t + 1$, seek a particular solution $p(t) = at + b$ led to $a = -at - b + 2t + 1$. Identify the coefficients given $a = 2$, $b = -1$. If y is a solution, $z = y - p$ checks $z' = -z$. With $y(0) = 3$, we get $z(0) = 4$, so $y(t) = 2t - 1 + 4e^{-t}$.

A4 · Partial guidance Solve $y' = 2y - 4$ with $y(0) = 3$ by subtracting a constant solution. Check the formula and initial value.

A5 · Repair Evidence One student checked that a function satisfies a differential equation, and then "I have found all the solutions." Why is this conclusion insufficient? Illustrate with $y' = y$.

7.5 Discrete, Continuous and Approximative Comparison

First use: S55 and S58.

Example Solved · Cooling

The model $T' = -0{,}1(T - 20)$, $T(0) = 80$ t in minutes, gives $T(t) = 20 + 60e^{-0{,}1t}$. With a step h, Euler gives $T_{n+1} = T_n - 0{,}1h(T_n - 20)$ in moments $t_n = nh$. For $h = 2$, on obtains 80,68,58,4. These are approximations of the continuous model at the moments 0.2,4; a law discrete physical given independently is not automatically Euler.

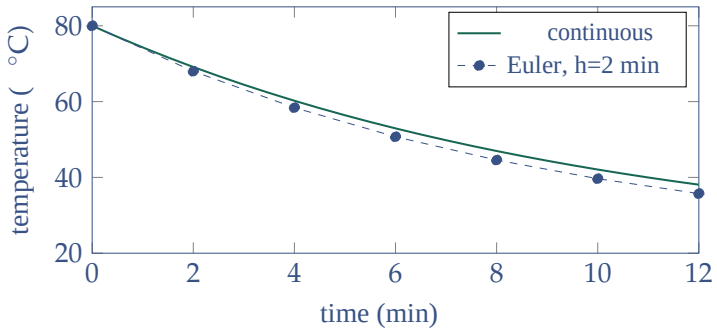


Figure 7.2. Read the deviation at t=6 min; segments connect successive approximations.

A6 · Choice of strategy Determine the first moment the continuous model reaches 50 degrees Celsius. Compare to the first value for Euler, with h=2 min, which is less than or equal to 50.
why the two results differ.

A first step: First solve an exponential equation; to Euler, write the formula of $T_n - 20$.

Wait a minute at the courthouse. Say what you mean by it, y' and t, then verify that the units of both members a differential equation is compatible.

Review postponed · S78. Complete the table of antiderivatives with trigonometric functions and control compositional factors.

Research activity

Cooling a drink · 40 min.

A drink initially at 80 °C is placed in a room held at 20 °C. For $t \geq 0$, note $T(t)$ its temperature in degrees Celsius, t the time elapsed in minutes. We admit the model $T'(t) = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial problem and give expression to $T(t)$.

2. Study the direction of variation in T and determine its limit in $+\infty$. Interpret results.

3. Determine the first time the temperature is 40 °C. Give the exact value, then a value approached in minutes.

4. Apply two steps of the Euler method with a step of 1 minute. Compare the approximation obtained after two minutes at the exact value $T(2)$.

5. Indicate a concrete situation in which the hypothesis of a constant ambient temperature would no longer be suitable.

Progressive exercises

E1 · Direct application. Determine all antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Application and justification. Determine the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ who verifies $F(0) = 0$.

E3 · Method to be organised. Determine the solution on $\mathbb{R}$ differential equation $y' = 3y$ who checks $y(1) = 2$.

E4 · Choice of strategy. Determine the solution on $\mathbb{R}$ differential equation $y' = -y + 4$ which checks $y(0) = 7$.

E5 · Synthesis or demonstration. The differential equation is considered $(E) : y' = y - x$ on $\mathbb{R}$. Verify that $g(x) = x + 1$ is a solution, then determine the solution of (E) who verifies $y(0) = 0$.

Work in the manual or annal

Barbazo, p. 228, Exercises 25 and 30. In S55: Exercise 30, to reveal a constant solution.
At home: Exercise 25 in full, distinguishing a condition on f a condition on f'.

Duty at home

DM7 · An evolution with variable input · 60 min.

We're looking for a function y defined and derivable on $[0; +\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Determine two actuals a and b so that $g(t) = at + b$ or a differential equation solution.

2. Deduce the expression of the solution that verifies $y(0) = 3$.

3. Study the variations of this solution and specify its minimum as well as the moment it is reached.

4. Write a Python argument function N, integer strictly positive, which applies the method d-Euler on $[0; 2]$ with the step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.

5. Why is the verification of differential equality in just a few moments not enough to prove that a function is a solution?

QCM A · antiderivative families

After S53; QCM B after S58.

For each question, only one proposal is accurate.
consult the correction.

1. A antiderivative of x^2 on $\mathbb{R}$ is:

A. $2x$ B. x^3

C. $x^3/3$ D. $3x^3$

2. All antiderivatives of e^{2x} are:

A. $e^{2x} + C$ B. $\frac{1}{2}e^{2x} + C$

C. $2e^{2x} + C$ D. $xe^{2x} + C$

3. A antiderivative of $2x/(1 + x^2)$ on $\mathbb{R}$ is:

A. $\frac{1}{1+x^2}$ B. $2\ln x$

C. $\ln x$ D. $\ln(1 + x^2)$

4. Two antiderivatives of the same function over an interval have curves:

A. Images of one another by vertical translation B. Necessarily identical

C. Always stinging D. Images by any horizontal translation

5. The antiderivative of $2x$ verifier $F(1) = 3$ is:

A. $x^2 + 3$ B. $2x^2 + 1$

C. $x^2 + 2$ D. $x^2 - 2$

QCM B · Differential Equation and Model

For each question, only one proposal is accurate.
consult the correction.

6. The solution of $y' = -2y$, $y(0) = 3$, is:

A. $3e^{2t}$ B. $3e^{-2t}$

C. $-2e^{3t}$ D. $3 - 2t$

7. The constant solution of $y' = -2y + 6$ is:

A. 6 B. -3

C. 2 D. 3

8. If y_p is a solution of $y' = ay + b(t)$, then $z = y - y_p$ verify:

A. $z' = az$ B. $z' = az + b(t)$

C. $z' = b(t)$ D. $z' = 0$ necessarily

9. Step by step h for $y' = -y$, from $y_0 = 2$, gives:

A. $y_1 = 2 + h$ B. $y_1 = e^{-h}$

C. $y_1 = 2 - 2h$ D. $y_1 = 2 - h^2$

10. If t is in minutes and T in degrees Celsius, the coefficient k of $T' = -k(T - 20)$ expressed in:

A. minutes B. Inverse minutes

C. degrees Celsius D. litres

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A antiderivative is unique without additional conditions.

2. Euler gives the exact value for any differential equation.

3. Checking a solution consists of checking the equation and the initial condition.

Corrected MCQs

1. **C.** Dirty $x^3/3$ give back x^2. Derivating x^2 is not looking for a antiderivative.

2. **B.** The factor 1/2 compensates for factor 2 of the compound bypass. C is a real constant.

3. **D.** It's the form u'/u, $u = 1 + x^2 > 0$ everywhere. x is not defined on all $\mathbb{R}$.

4. **A.** The difference between antiderivatives is constant; this translates into a vertical translation.

5. **C.** The antiderivatives are $x^2 + C$. The condition gives $1 + C = 3$, so $C = 2$.

6. **B.** The solutions are Ce^{-2t} ; the initial condition gives $C = 3$. An affine function is not suitable Every moment.

7. **D.** A constant has a null derivative; solve $0 = -2L + 6$ given $L = 3$.
8. **A.** Subtracting the two equations eliminates the second member. This is the associated homogeneous equation.
9. **C.** The update is $y_1 = y_0 + h(-y_0) = 2 - 2h$. Euler gives an approximation, distinct from the exact solution $2e^{-h}$.
10. **B.** The derivative is in degrees Celsius per minute: k multiplies a temperature difference and a For unit min^{-1}.

True/false. False: adding a constant gives another antiderivative. False: for $y' = -y$, $y(0) = 2$, Step 1 gives 0, instead of $2/e$. True, on the required interval and with the required differentiability.

Corrected applications and support

Application 1. $x^4 - x^2 + x$; $-e^{-3x}/3$; $\frac{1}{2}\ln|2x+1|$ on each of the intervals not containing $-1/2$. Derivatives give back the announced expressions.

Application 2. $F = x^2 + C$ and $F(1) = 3$ give C=2. On the two half-rights, $\ln|x| + C_-$ for $x < 0$ and $\ln|x| + C_+$ for $x > 0$ is a antiderivative, even if $C_- \neq C_+$.

Implementation 3. $y = 5e^{-2t}$. The null function satisfies the equation alone, but not the condition $y(0) = 5$.

Implementation 4. Constant 2 is a solution. $z = y - 2$ check $z' = 2z$ and $z(0) = 1$, therefore $y = 2 + e^{2t}$. Its derivative $2e^{2t}$ is good 2 $y - 4$ and $y(0) = 3$.

Application 5. Check e^t shows only that it is a solution. Functions 2 e^t, $-e^t$ and the for completeness, starting from any solution and establishing that it a the form announced.

Implementation 6. Ongoing: $e^{-0{,}1t} = 1/2$, or $t = 10\ln 2 \simeq 6{,}93$ min. Euler: $T_n = 20 + 60(0{,}8)^n$; $(0{,}8)^3 = 0{,}512 > 1/2$, $(0{,}8)^4 = 0{,}4096 < 1/2$. The first point of the grid is therefore 8 min. imposed discrete moments and Euler introduced an error of approximation.

Research and comparison

Calculate a first step by hand, then compare h=2 and h=1 with the program provided.
The observed convergence of an approximation does not replace an estimate of its error.
Do not call "exact solution" a list of Euler values.

Conduct and correction

Benchmarks for the teacher

Distinguish unknown function and actual variable. Have each antiderivative checked by bypass. For $y' = ay + b$, first seek the constant solution. The case $a = 0$ must remain visible. Trigonometric antiderivatives are given here as reference table, but their use Evaluated waits for the trigonometry sequence. To Euler, keep the units and compare to a known exact solution.

Corrected activity

1. The equation is $T' = -0{,}1T + 2$. A constant solution is 20; all solutions are $T(t) = 20 + Ce^{-0{,}1t}$. Like $T(0) = 80$, $C = 60$. Verification $T'(t) = -6e^{-0{,}1t} = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

2. $T'(t) = -6e^{-0{,}1t} < 0$: temperature decreases strictly. $e^{-0{,}1t} \to 0$ in $+\infty$, $T(t) \to 20\,°\text{C}$, remaining above 20 °C.

3. $T(t) = 40 \iff e^{-0{,}1t} = \frac{1}{3}$. Taking the logarithm, $-0{,}1t = -\ln 3$, either $t = 10\ln 3 \simeq 10{,}9861$ min. Strict decline ensures that it is the only moment of transition to 40 °C.

4. For $h = 1$, $Y_{k+1} = Y_k - 0{,}1(Y_k - 20)$. Thus $Y_1 = 80 - 6 = 74$ and $Y_2 = 74 - 5{,}4 = 68{,}6$. The solution exact given $T(2) = 20 + 60e^{-0{,}2} \simeq 69{,}1238$. To underestimate here the temperature of about 0,5238 °C.

5. The model assumes an ambient temperature of 20 °C and a constant exchange coefficient of $0{,}1\ \text{min}^{-1}$. A change of part, lid or agitation may alter these assumptions; the formula is no longer automatically adapted.

E1. Linearity gives $F(x) = x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. We're checking. $F'(x) = 3x^2 - 4x + 1$. All antiderivatives on the interval $\mathbb{R}$ different from a constant.

E2. The denominator is positive and its derivative is 2 x. The antiderivatives are $\ln(x^2 + 4) + C$. The condition in 0 gives $C = -$ No 4, therefore. $F(x) = \ln(x^2 + 4) - 4$; check $F(0) = 0$ and $F'(x) = \frac{2x}{x^2+4}$.

E3. The solutions of $y' = 3y$ are Ce^{3x}. The condition $Ce^3 = 2$ requires $C = 2e^{-3}$. Thus $y(x) = 2e^{3(x-1)}$, of which the derivative is $3y(x)$ and the value in 1 is 2.

E4. Constant 4 is a particular solution, so the solutions are 4 $+ Ce^{-x}$. Like 4 $+ C = 7$, $C = 3$. The solution is $y(x) = 4 + 3e^{-x}$; its derivative $-3e^{-x}$ is good $-y + 4$, and $y(0) = 7$.

E5. $g(x) = x + 1$ check $g' = 1 = g - x$. If $z = y - g$, then $z' = z$, therefore $y = x + 1 + Ce^x$. The condition $y(0) = 0$ tax $C = -1$. One obtains $y = x + 1 - e^x$ and we're checking $y' = 1 - e^x = y - x$.

Corrected from DM7

1. If $g(t) = at + b$, then $g' = a$ and $-g + 2t + 1 = (2 - a)t + 1 - b$. For identity to be true to everything t, it takes $2 - a = 0$ and $a = 1 - b$. So $a = 2$, $b = -1$, and $g(t) = 2t - 1$.

2. Poisons $z = y - g$. By subtracting the two equations, $z' = -z$, therefore $z = Ce^{-t}$. Thus $y(t) = 2t - 1 + Ce^{-t}$. Like $y(0) = 3$, $C = 4$. The function $y(t) = 2t - 1 + 4e^{-t}$ Check it out. $y' = 2 - 4e^{-t} = -y + 2t + 1$ and $y(0) = 3$.

3. $y'(t) = 2 - 4e^{-t}$. This derivative is negative for $t < \ln 2$, none in $\ln 2$, positive after. The solution decreases on $[0, \ln 2]$ then grows on $[\ln 2, +\infty[$. Its minimum is $y(\ln 2) = 2\ln 2 - 1 + 2 = 1 + 2\ln 2$.

4.
```
def euler_dm7(N):
    h = 2/N
    y = 3.0
    for k in rows (N):
        t = k*h
        y = y+h*(-y+2*t+1)
    return y
```

The time used in the second member is that of the beginning of the step.
$Y_{10} \simeq 3{,}42950$ and $Y_{100} \simeq 3{,}53048$, while $y(2) = 3 + 4e^{-2} \simeq 3{,}54134$. Absolute errors are valid about 0 ,11184 and 0,01086. To verify these values without loop, posisons $Z_k = Y_k - (2kh - 1)$. The recurrence of Euler gives $Z_{k+1} = (1-h)Z_k$ and $Z_0 = 4$. Therefore $Y_N = 3 + 4(1 - \frac{2}{N})^N$.

5. A differential equation solution shall check for equality of functions throughout the interval, and the initial condition. Some numerical values control only a finite number of
The symbolic substitution of question 2 is a verification of the whole field.

Corrected work in the manual or an anal

Exercise 30. If y is constant equal to ℓ, the equation becomes $0 = -\frac{1}{2}\ell + 3$, so $\ell = 6$. By posing $z = y - 6$, we get $z' = -\frac{1}{2}z$; all solutions are $y(x) = 6 + Ce^{-\frac{x}{2}}$, $C \in \mathbb{R}$.

Exercise 25.

1. $f' = \frac{1}{2}f$ gives $f(x) = Ce^{\frac{x}{2}}$. Like $f(0) = 1$, $C = 1$.

2. $5f' = f$ gives $f(x) = Ce^{\frac{x}{5}}$. The condition $f(1) = 0$ tax $Ce^{\frac{1}{5}} = 0$; the exponential being never None, $C = 0$. The solution is the null function.

3. $f' + f = 0$ given $f(x) = Ce^{-x}$, therefore $f'(x) = -Ce^{-x}$. The condition relates to the derivative: $f'(1) = 2$ is equivalent to $-Ce^{-1} = 2$, or $C = -2e$. Thus $f(x) = -2e^{1-x}$.